

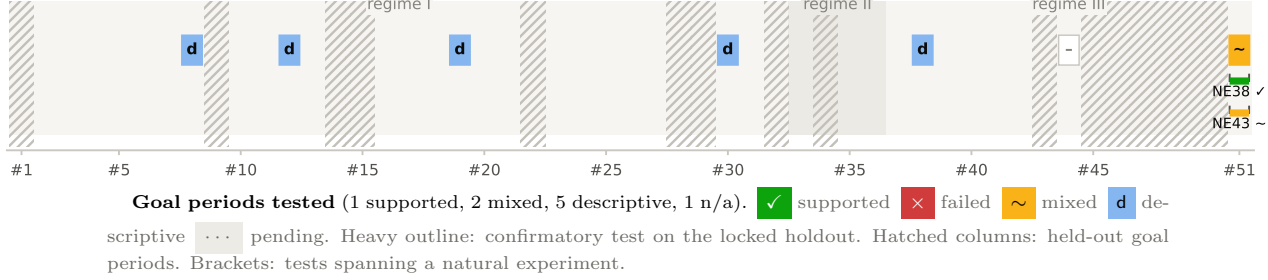
H104: Barkhausen avalanches when a field is stepped

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

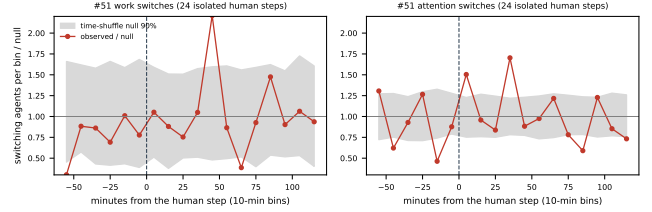
Bottom line. Human messages do not release switch avalanches in the AI Village: the step response is within 10% of the null where it can be measured, and the crackling tail cannot be identified with fewer than 25 steps.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH330 proposed Barkhausen crackling: when a human steps the external field mid-period, agents switch projects in avalanches with a heavy-tailed size law ($\tau \approx 3/2$), and between steps they switch at random. The test uses work switches (DQ4) and attention switches (project states) on 8 eligible non-holdout period-channels. Only #51 has power (24 isolated human steps). There, switching in the hour after a step is $\times 1.09$ [0.84, 1.32] (work) and $\times 1.07$ [0.97, 1.18] (attention) the time-shuffle null. A planted $\tau = 3/2$ avalanche gives $\times 1.87$ and $\times 1.25$. Big bursts are as frequent between steps as after them. Between steps, switching is near-Poisson given each agent's rate. Kickoffs do move agents. **The claim fails**; the tail exponent is not identifiable at village counts. The holdout is not run.



Observables. Distinct agents switching per 10-min bin around the 24 isolated human steps in #51, divided by the time-shuffle null (each agent-day's switches rotated within its at-risk span). Gray: the null's 90% band. The curve stays inside the band in both channels. Single bins outside it (+45 min work, +5 and +35 min attention) are about what 18 bins give by chance.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator, a mid-period message is a weak lever on what a swarm works on. It moves the addressed agent (NE38) but does not set off a cascade: the upper bound on the response is 18–32% more switching in the next hour. To redirect a whole swarm, use a kickoff-scale change: on kickoff days 48–100% of agents switch in the first 2 h, against 17–86% on other days. For a monitor, switching between interventions is close to Poisson given each agent's own rate ($D \approx 1.1$ –1.2), so an unusual cluster of switches is a real event. Crackling exponents need far more steps than the village has. Do not fit them to an LLM swarm's logs at this scale.

PHYSICAL MODEL

Zero-temperature mean-field random-field Ising/Potts model driven by field steps (models 01, 10, 14). Each agent is a Potts spin σ_i (its current project) with a quenched field h_i and a coupling J to the swarm. A human session at t_k steps the field by ΔH_k . Spins that cross threshold flip and can trigger others: an avalanche of A_k flips. The burst size S_k adds background flips B_k , whose distribution the time shuffle supplies.

$$P(A) \propto A^{-\tau} \mathcal{D}(A/A^*), \quad \tau = \frac{3}{2} \text{ (mean-field RFIM)}, \quad S_k = A_k + B_k$$

$$X = \frac{\langle S_k \rangle}{\langle S_k^{\text{null}} \rangle}, \quad F_A = \frac{\text{Var } S - \text{Var } S^{\text{null}}}{\langle S \rangle - \langle S^{\text{null}} \rangle}, \quad D = \frac{\text{Var } S_{\text{quiet}}}{\text{Var } S_{\text{null}}^{\text{quiet}}}$$

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RESULTS VS PREDICTIONS

FAITHFULNESS SCORECARD

Prediction (dated)	Observed	Verdict
P1 steps release switches ($X > 1$)	pooled attention 0.91 [0.72, 1.14], work 1.04 [0.86, 1.25]; #51 1.07, 1.09 (CIs incl. 1)	failed
P2 heavy tail, $\tau \in [1.2, 1.8]$	V 1.02, 1.19; F_A CIs incl. 1; no avalanche component	inconclusive (A1)
P3 Poisson between steps; BR > 1	#51 D 1.12, 1.18; BR CIs incl. 1	D yes; BR failed
P4 burst grows with step size	ρ -0.30, -0.10 (n.s.)	failed
P5 kickoffs give the largest bursts	$K > 1$ in 7/8 (1.16–3.97)	supported
P7 no reverse causation	pre-step X 0.82, 0.93	supported
NE38 local step stays local	target switches; others inside band	supported
NE43 nudger stop	ratio 1.22 [0.72, 2.03], 0.96 [0.86, 1.07]; inside placebo	mixed
G44 room-local sponse	no response to localize	n/a

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	1		
Level: hypothesis . Scale 0 not done or failed, 1 partial, 2 passed (<code>writeup/paper.tex</code>).			

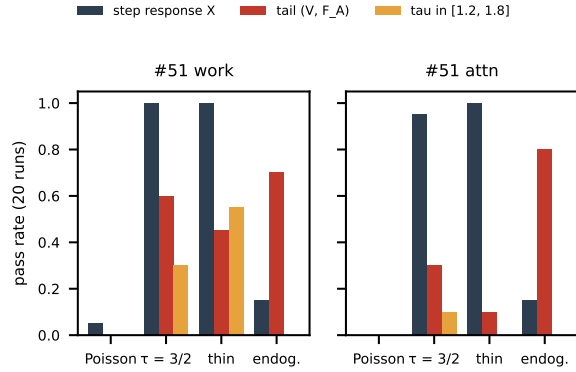
CAVEATS

Only #51 has power; the other periods have 3–4 isolated steps. Attention switches have 15-min resolution. The at-risk trim removes each agent's first 30 min, so morning steps rarely qualify. Work switches need a commit, so idle agents cannot switch, and the read-out split is uninformative. The step size (message count) is a crude field magnitude. Single variants show at most a 10–20% response (switch counts: attention $\times 1.13$, p 0.013). Predictions were written knowing H28, H53 and H63.

ROUND 2

Event-study the kickoffs as the real field steps. Restrict human steps to messages that name a project. Move the spin to 30-min content states, which flip far more often than projects and may give tail power.

- **Essence.** which inputs move a swarm's allocation of effort, and by how much.
- **H104-R1** Kickoff-scale steps.
- **H104-R2** Directed vs broadcast human steps.
- **H104-R3** Avalanches in content, not projects.



Second figure. Synthetic validation at #51 counts (20 runs per world). The step response X separates a response from none. The tail criterion ($V > 1$ and F_A CI > 1) and the τ window pass as often for a thin Poisson response, or for endogenous bursts, as for $\tau = 3/2$ avalanches. The tail is not identifiable at 24 steps.

GOAL PERIODS IN DETAIL

G08 d descriptive
G12 d descriptive
G19 d descriptive
G30 d descriptive
G38 d descriptive
G51 ~ mixed
NE38 ✓ supported
NE43 ~ mixed
G44 - n/a

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